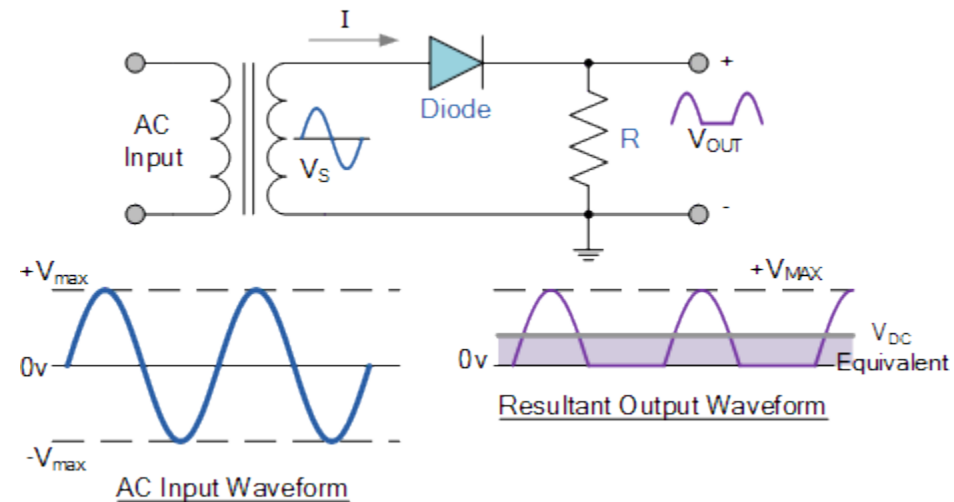


Diode Applications

- Diodes and Rectifiers

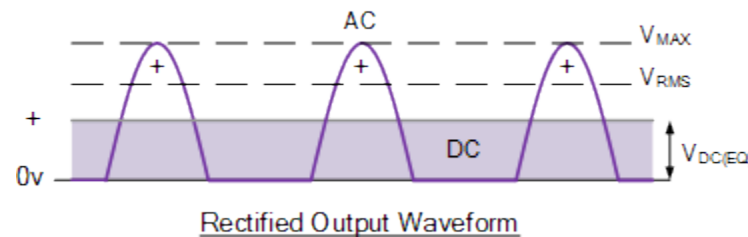
- A widely used application of this feature and diodes in general is in the conversion of an alternating voltage (AC) into a continuous voltage (DC). In other words, Rectification.
- If an alternating voltage is applied across a diode, during the positive half cycle the diode will conduct passing current and during the negative half cycle the diode will not conduct blocking the flow of current. Then conduction through the power diode only occurs during the positive half cycle and is therefore unidirectional i.e. DC.

- Half Wave Rectifier



- The power diode in a half wave rectifier circuit passes just one half of each complete sine wave of the AC supply in order to convert it into a DC supply. Then this type of circuit is called a “half-wave” rectifier because it passes only half of the incoming AC power supply

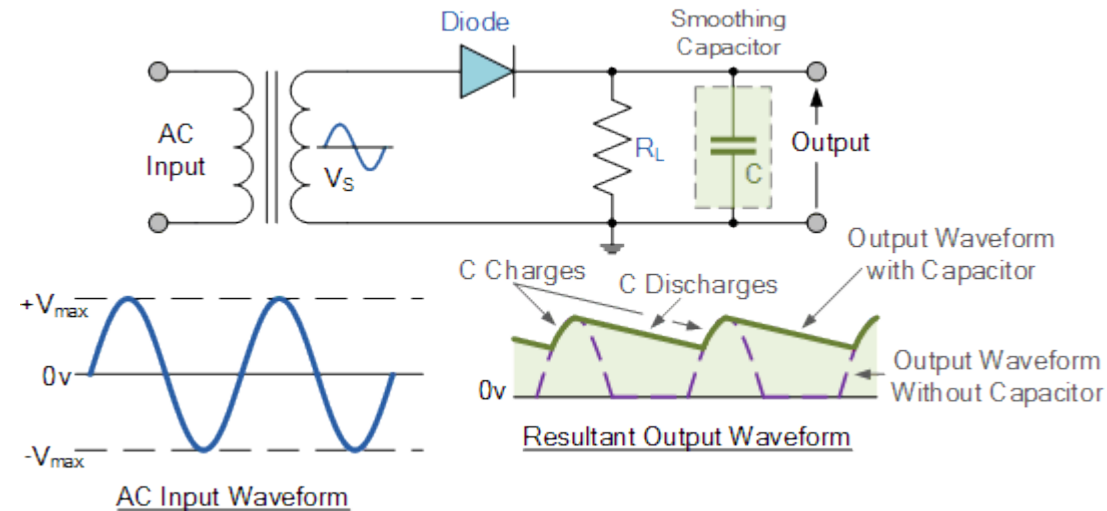
- Since the DC load is resistive (resistor, R), the current flowing in the load resistor is therefore proportional to the voltage (Ohm's Law), and the voltage across the load resistor will therefore be the same as the supply voltage, V_s , that is the "DC" voltage across the load is sinusoidal for the first half cycle only so $V_{out} = V_s$.
- During each "negative" half cycle of the AC sinusoidal input waveform, the diode is reverse biased as the anode is negative with respect to the cathode. Therefore, NO current flows through the diode or circuit. Then in the negative half cycle of the supply, no current flows in the load resistor as no voltage appears across it so therefore, $V_{out} = 0$.
- The current on the DC side of the circuit flows in one direction only making the circuit Unidirectional. As the load resistor receives from the diode a positive half of the waveform, zero volts, a positive half of the waveform, zero volts, etc, the value of this irregular voltage would be equal in value to an equivalent DC voltage of $0.318 \cdot V_{max}$ of the input sinusoidal waveform or $0.45 \cdot V_{rms}$ of the input sinusoidal waveform.



- $V_{dc} = \frac{V_{max}}{\pi} = 0.318V_{max} = 0.45V_{rms}, \quad (V_{rms} = 0.707V_{max})$
- Where V_{MAX} is the maximum or peak voltage value of the AC sinusoidal supply, and V_{RMS} is the RMS (Root Mean Squared) value of the supply voltage.

- Half-wave Rectifier with Smoothing Capacitor

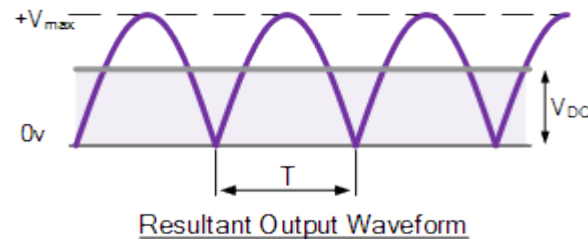
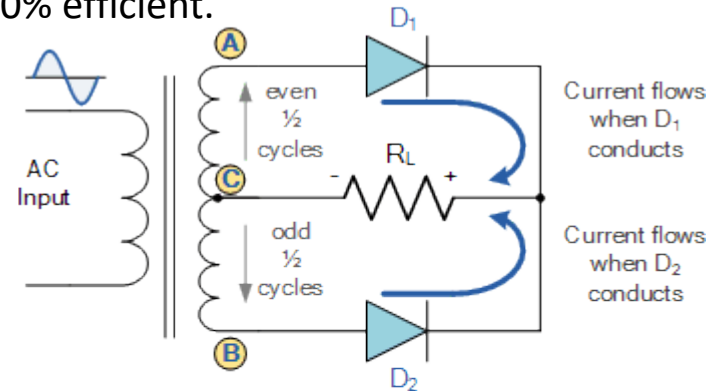
- When rectification is used to provide a direct voltage (DC) power supply from an alternating (AC) source, the amount of ripple voltage can be further reduced by using larger value capacitors but there are limits both on cost and size to the types of smoothing capacitors used.
- For a given capacitor value, a greater load current (smaller load resistance) will discharge the capacitor more quickly (RC Time Constant) and so increases the ripple obtained. Then for single phase, half-wave rectifier circuit using a power diode it is not very practical to try and reduce the ripple voltage by capacitor smoothing alone.



- In practice, the half-wave rectifier is used most often in low-power applications because of their major disadvantages being. The output amplitude is less than the input amplitude, there is no output during the negative half cycle so half the power is wasted and the output is pulsed DC resulting in excessive ripple.
- In this instance it would be more practical to use "Full-wave Rectification" instead.

- Full Wave Rectifier

- In a Full Wave Rectifier circuit two diodes are now used, one for each half of the cycle. A multiple winding transformer is used whose secondary winding is split equally into two halves with a common center tapped connection, (C). This configuration results in each diode conducting in turn when its anode terminal is positive with respect to the transformer center point C producing an output during both half-cycles, twice that for the half wave rectifier so it is 100% efficient.

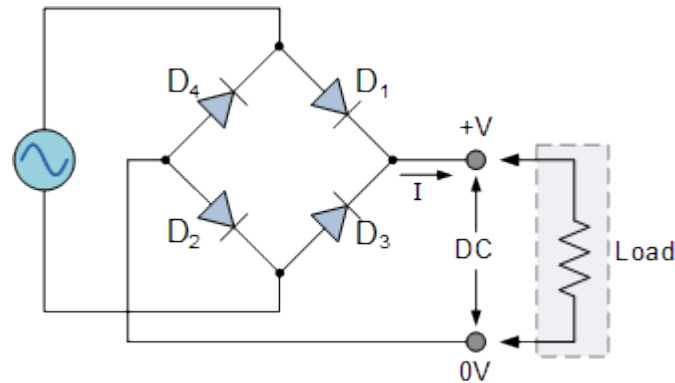


$$V_{dc} = \frac{2V_{max}}{\pi} = 0.637V_{max} = 0.9V_{rms}$$

- The full wave rectifier circuit consists of two diodes connected to a single load resistance (RL) with each diode taking it in turn to supply current to the load. When point A of the transformer is positive with respect to point C, diode D1 conducts in the forward direction as indicated by the arrows.
- When point B is positive (in the negative half of the cycle) with respect to point C, diode D2 conducts in the forward direction and the current flowing through resistor R is in the same direction for both half-cycles. As the output voltage across the resistor R is the phasor sum of the two waveforms combined, this type of full wave rectifier circuit is also known as a “bi-phase” circuit.

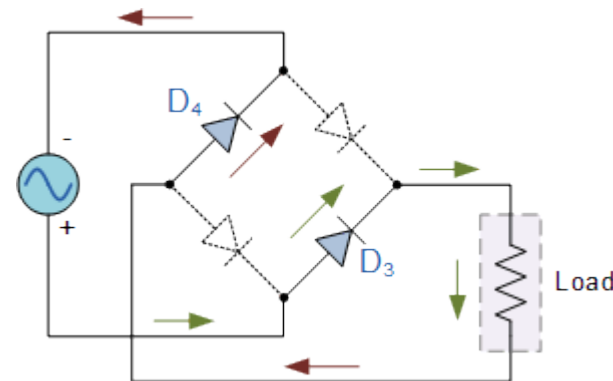
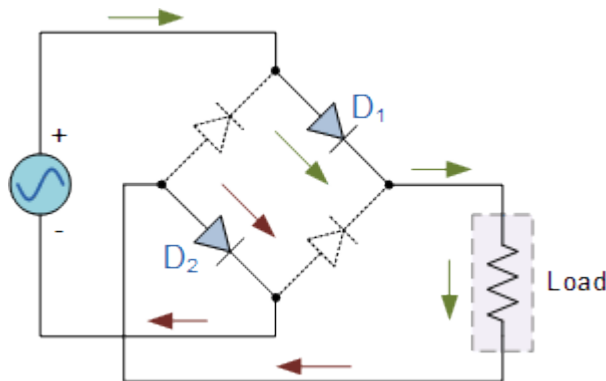
- The Full Wave Bridge Rectifier

- Another type of circuit that produces the same output waveform as the full wave rectifier circuit above, is that of the Full Wave Bridge Rectifier. This type of single phase rectifier uses four individual rectifying diodes connected in a closed loop “bridge” configuration to produce the desired output.
- The main advantage of this bridge circuit is that it does not require a special center tapped transformer, thereby reducing its size and cost. The single secondary winding is connected to one side of the diode bridge network and the load to the other side.



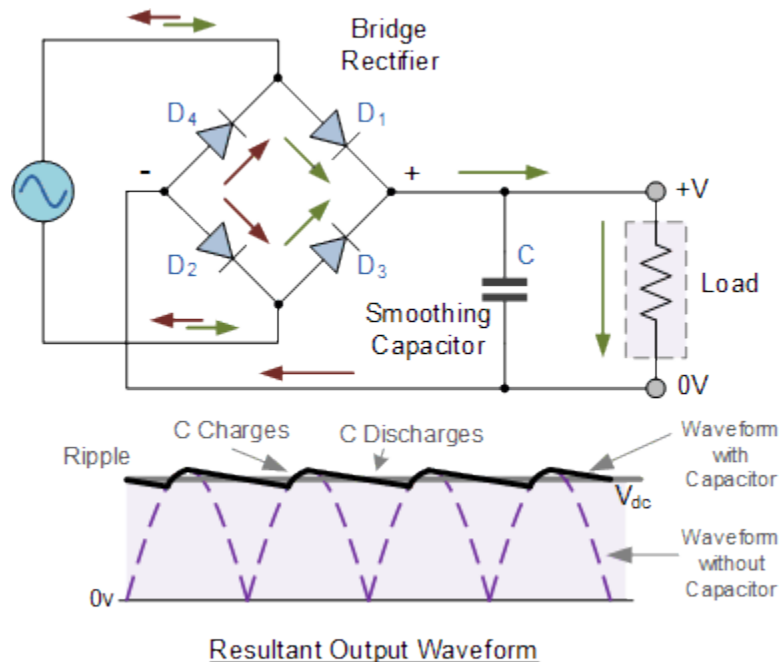
- The average DC voltage across the load is $0.637V_{\max}$.
- However in reality, during each half cycle the current flows through two diodes instead of just one so the amplitude of the output voltage is two voltage drops ($2 \times 0.7 = 1.4V$) less than the input $V_{\max}$ amplitude.

- The four diodes labelled D_1 to D_4 are arranged in “series pairs” with only two diodes conducting current during each half cycle. During the positive half cycle of the supply, diodes D_1 and D_2 conduct in series while diodes D_3 and D_4 are reverse biased and the current flows through the load as shown below (Opposite in the negative cycle).



- Full-wave Rectifier with Smoothing Capacitor

- The single phase half-wave rectifier produces an output wave every half cycle and that it was not practical to use this type of circuit to produce a steady DC supply. The full-wave bridge rectifier however, gives us a greater mean DC value ($0.637 V_{\max}$) with less superimposed ripple while the output waveform is twice that of the frequency of the input supply frequency.
- We can improve the average DC output of the rectifier while at the same time reducing the AC variation of the rectified output by using smoothing capacitors to filter the output waveform. Smoothing or reservoir capacitors connected in parallel with the load across the output of the full wave bridge rectifier circuit increases the average DC output level even higher as the capacitor acts like a storage device



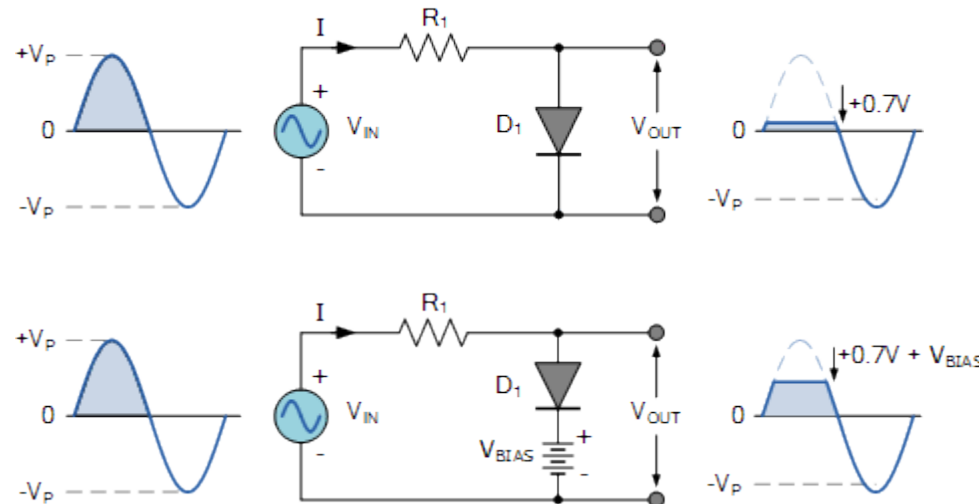
- $V_{ripple} = \frac{I_{load}}{f \times C}$
- Where: I is the DC load current in amps, f is the frequency of the ripple or twice the input frequency in Hertz, and C is the capacitance in Farads.

- Diode Clipping Circuits

- This clipping of the input signal produces an output waveform that resembles a flattened version of the input. For example, the half-wave rectifier is a clipper circuit, since all voltages below zero are eliminated.

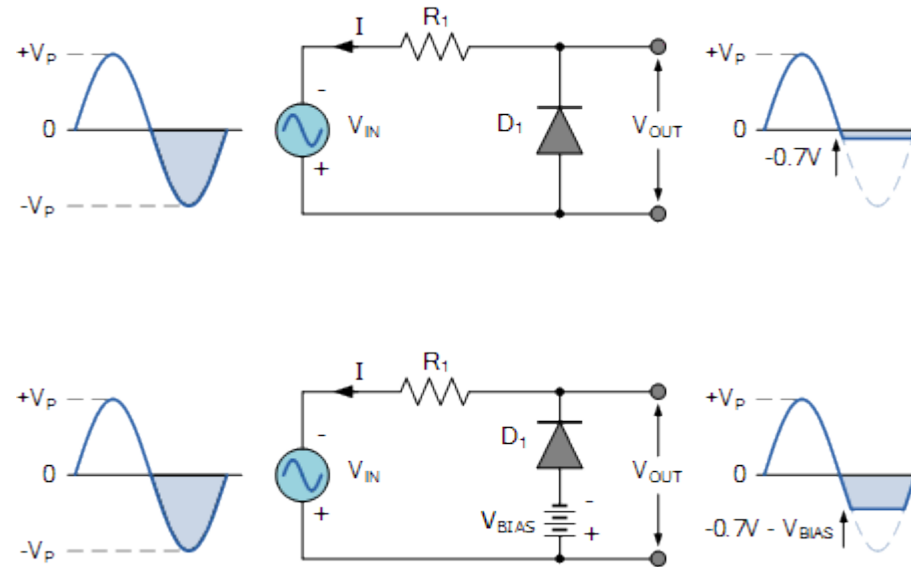
- Positive Diode Clipping Circuits

- In this diode clipping circuit, the diode is forward biased (anode more positive than cathode) during the positive half cycle of the sinusoidal input waveform. For the diode to become forward biased, it must have the input voltage magnitude greater than +0.7 volts (0.3 volts for a germanium diode).
- When this happens the diodes begins to conduct and holds the voltage across itself constant at 0.7V until the sinusoidal waveform falls below this value. Thus the output voltage which is taken across the diode can never exceed 0.7 volts during the positive half cycle.
- During the negative half cycle, the diode is reverse biased (cathode more positive than anode) blocking current flow through itself and as a result has no effect on the negative half of the sinusoidal voltage which passes to the load unaltered. Thus the diode limits the positive half of the input waveform and is known as a positive clipper circuit.



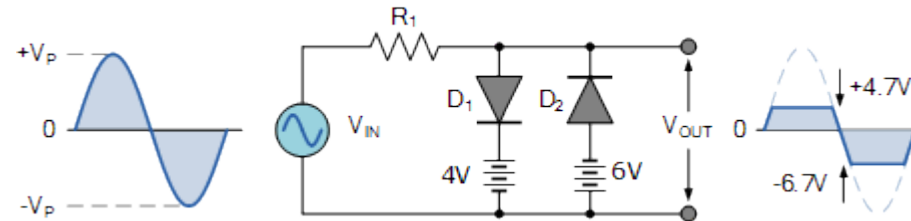
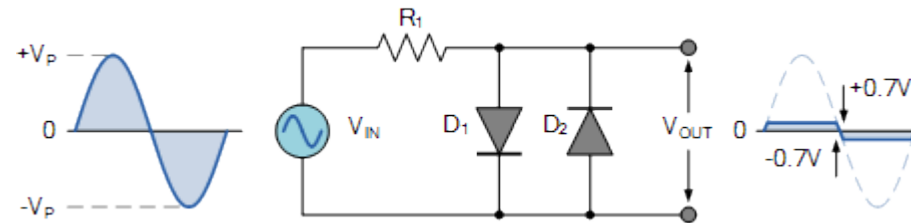
- Negative Diode Clipping Circuits

- The diode is forward biased during the negative half cycle of the sinusoidal waveform and limits or clips it to -0.7 volts while allowing the positive half cycle to pass unaltered when reverse biased. As the diode limits the negative half cycle of the input voltage it is therefore called a negative clipper circuit.



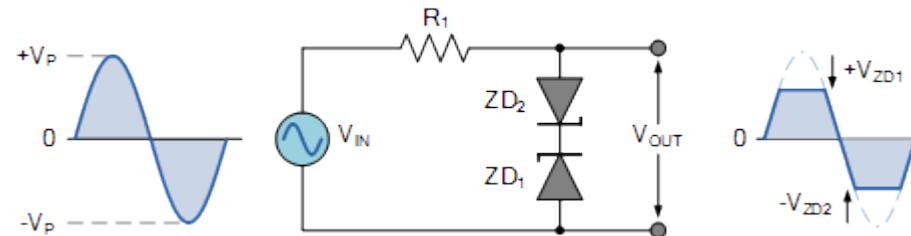
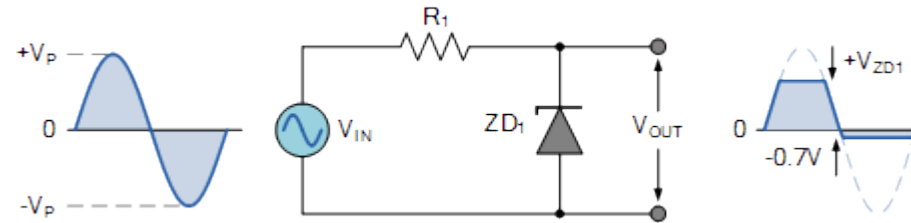
- Clipping of Both Half Cycles

- If we connected two diodes in inverse parallel as shown, then both the positive and negative half cycles would be clipped as diode D_1 clips the positive half cycle of the sinusoidal input waveform while diode D_2 clips the negative half cycle. Then diode clipping circuits can be used to clip the positive half cycle, the negative half cycle or both.
- For ideal diodes the output waveform above would be zero. However, due to the forward bias voltage drop across the diodes the actual clipping point occurs at $+0.7$ volts and -0.7 volts respectively. But we can increase this $\pm 0.7V$ threshold to any value we want up to the maximum value, (V_{PEAK}) of the sinusoidal waveform either by connecting together more diodes in series creating multiples of 0.7 volts, or by adding a voltage bias to the diodes.



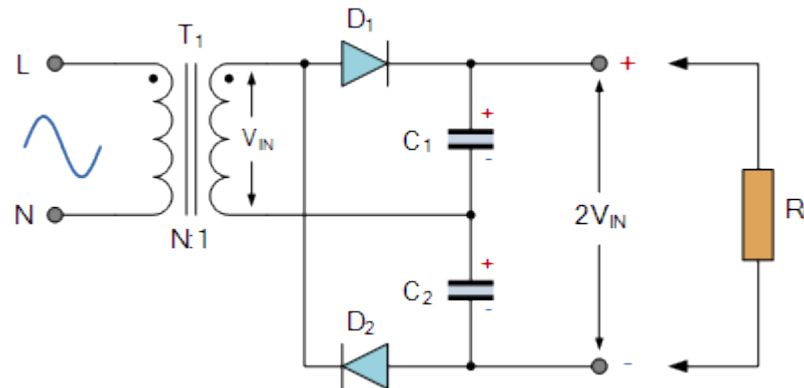
- Zener Diode Clipping

- The Zener diode is acting like a biased diode clipping circuit with the bias voltage being equal to the Zener breakdown voltage. In this circuit during the positive half of the waveform the Zener diode is reverse biased so the waveform is clipped at the Zener voltage, V_{ZD1} . During the negative half cycle the Zener acts like a normal diode with its usual 0.7V junction value.



- Voltage Multiplier

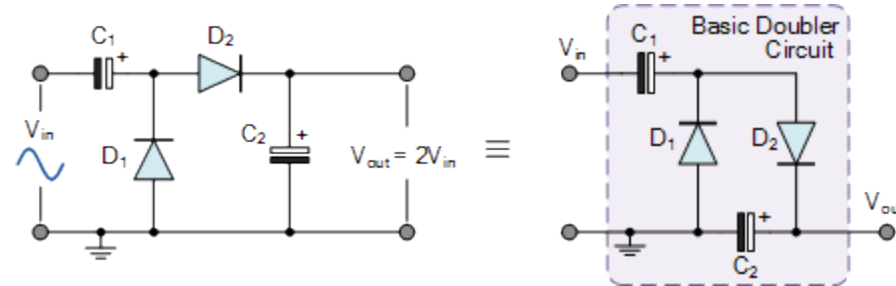
- Voltage multipliers are similar in many ways to rectifiers in that they convert AC-to-DC voltages for use in many electrical and electronic circuit applications such as in microwave ovens, strong electric field coils for cathode-ray tubes, electrostatic and high voltage test equipment, etc, where it is necessary to have a very high DC voltage generated from a relatively low AC supply.
- Generally, the DC output voltage (V_{dc}) of a rectifier circuit is limited by the peak value of its sinusoidal input voltage. But by using combinations of rectifier diodes and capacitors together we can effectively multiply this input peak voltage to give a DC output equal to some odd or even multiple of the peak voltage value of the AC input voltage.



- When the sinusoidal input voltage is positive, capacitor C_1 charges up through diode D_1 and when the sinusoidal voltage is negative, capacitor C_2 charges up through diode, D_2 . The output voltage $2V_{IN}$ is taken across the two series connected capacitors.
- The voltage produced by a voltage multiplier circuit is in theory unlimited, but due to their relatively poor voltage regulation and low current capability there are generally designed to increase the voltage by a factor less than ten. However, if designed correctly around a suitable transformer, voltage multiplier circuits are capable of producing output voltages in the range of a few hundred to tens of thousands of volts, depending upon their original input voltage value but all with low currents in the milliamperes range.

- The Voltage Doubler

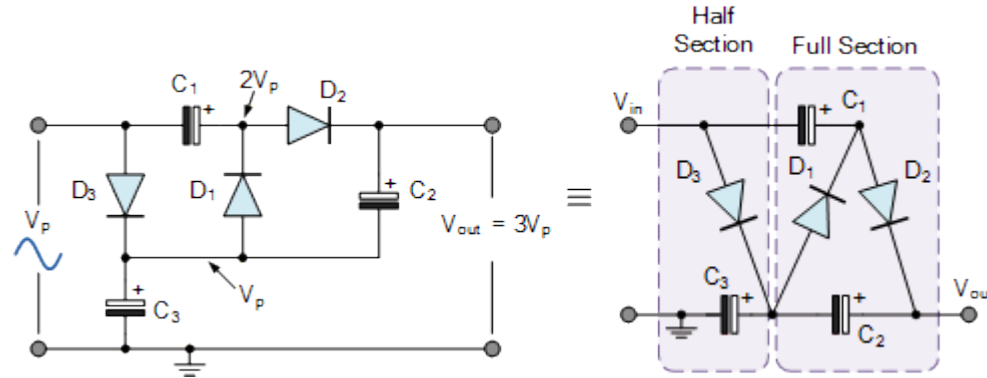
- A Voltage Doubler is a voltage multiplier circuit which has a voltage multiplication factor of two. The circuit consists of only two diodes, two capacitors and an oscillating AC input voltage. This simple diode-capacitor pump circuit gives a DC output voltage equal to the peak-to-peak value of the sinusoidal input. In other words, double the peak voltage value because the diodes and the capacitors work together to effectively double the voltage.



- The circuit shows a half wave voltage doubler. During the negative half cycle of the sinusoidal input waveform, diode D_1 is forward biased and conducts charging up the pump capacitor, C_1 to the peak value of the input voltage, (V_p). Because there is no return path for capacitor C_1 to discharge into, it remains fully charged acting as a storage device in series with the voltage supply. At the same time, diode D_2 conducts via D_1 charging up capacitor, C_2 .
- During the positive half cycle, diode D_1 is reverse biased blocking the discharging of C_1 while diode D_2 is forward biased charging up capacitor C_2 . But because there is a voltage across capacitor C_1 already equal to the peak input voltage, capacitor C_2 charges to twice the peak voltage value of the input signal.

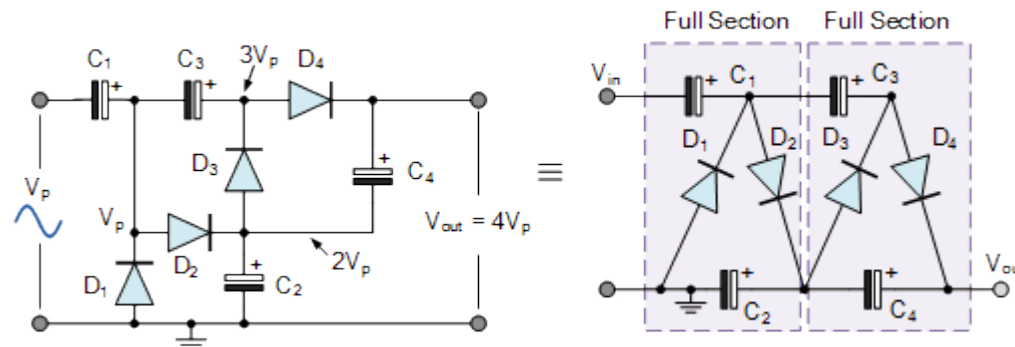
- DC Voltage Tripler Circuit

- By adding an additional single diode-capacitor stage to the half-wave voltage doubler circuit above, we can create another voltage multiplier circuit that increases its input voltage by a factor of three and producing what is called a Voltage Tripler Circuit.



- DC Voltage Quadrupler Circuit

- The first voltage multiplier stage doubles the peak input voltage and the second stage doubles it again, giving a DC output equal to four times the peak voltage value ($4V_p$) of the sinusoidal input signal. Also, using large value capacitors will help to reduce the ripple voltage.

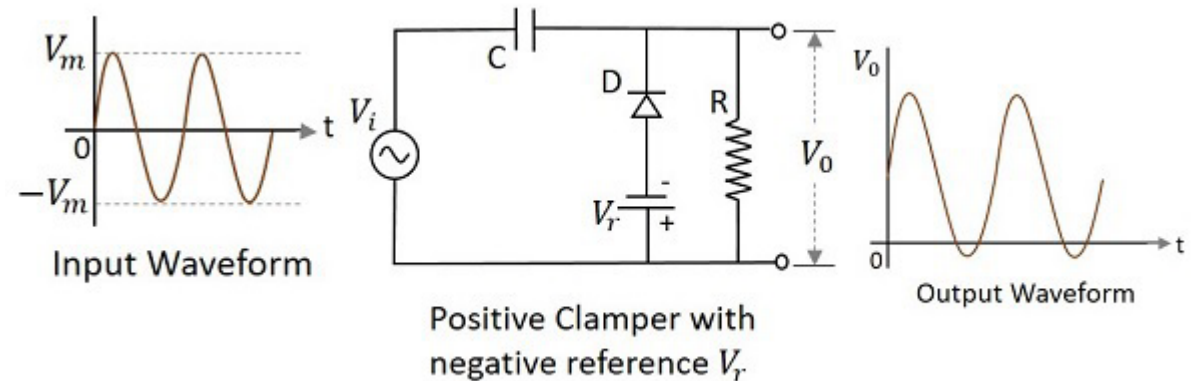
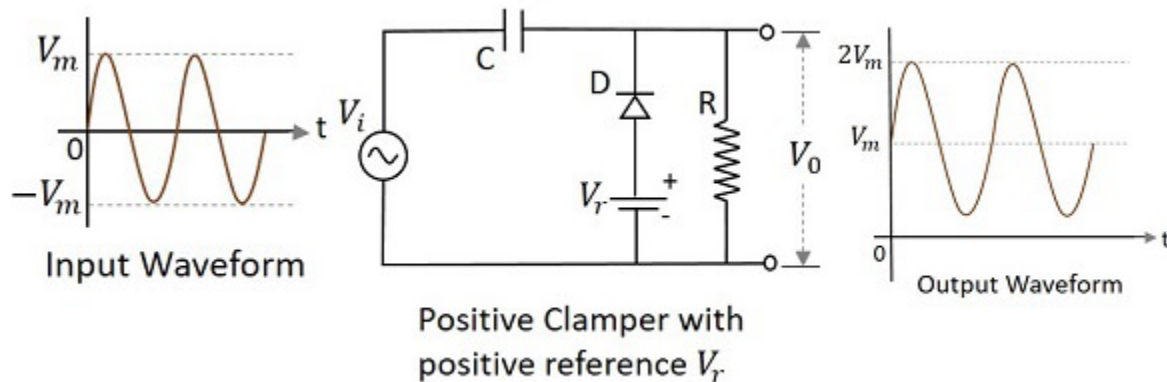
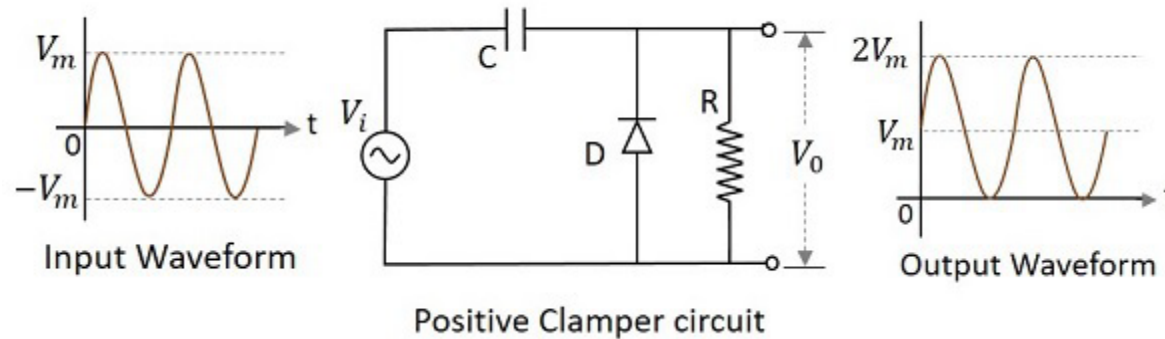


- Clamper Circuit

- A Clamper circuit can be defined as the circuit that consists of a diode, a resistor and a capacitor that shifts the waveform to a desired DC level without changing the actual appearance of the applied signal.

- Positive Clamper Circuit

- A Clamping circuit restores the DC level. When a negative peak of the signal is raised above to the zero level, then the signal is said to be positively clamped.



- Light Emitting Diode

- The “Light Emitting Diode” or LED as it is more commonly called, is basically just a specialized type of diode as they have very similar electrical characteristics to a PN junction diode. This means that an LED will pass current in its forward direction but block the flow of current in the reverse direction.
- Light emitting diodes are made from a very thin layer of fairly heavily doped semiconductor material and depending on the semiconductor material used and the amount of doping, when forward biased an LED will emit a coloured light at a particular spectral wavelength.
- When the diode is forward biased, electrons from the semiconductors conduction band recombine with holes from the valence band releasing sufficient energy to produce photons which emit a monochromatic (single colour) of light. Because of this thin layer a reasonable number of these photons can leave the junction and radiate away producing a coloured light output.

Typical LED Characteristics			
Material	Wavelength	Colour	V_F @ 20mA
GaAs	850-940nm	Infra-Red	1.2v
GaAsP	630-660nm	Red	1.8v
GaAsP	605-620nm	Amber	2.0v
GaAsP:N	585-595nm	Yellow	2.2v
AlGaP	550-570nm	Green	3.5v
SiC	430-505nm	Blue	3.6v
GaN	450nm	White	4.0v

